Musterwerk Automotive GmbH

Lastenheft: Adaptive Cruise Control (ACC)

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Revision History

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1.0	2026-10-28	B. Aksakal	Benchmark version with answer key

1 Introduction

This document specifies the Adaptive Cruise Control (ACC) of passenger cars (category M1). ACC keeps the speed set by the driver and adapts it to the vehicle ahead. Requirements are identified by IDs like ACC-R-010.

2 Speed Control

ACC-R-010 Set speed range

The driver shall be able to set a target speed between 30 km/h and 160 km/h.

ACC-R-011 Maximum set speed

The maximum target speed the driver can set shall be 180 km/h.

ACC-R-012 Minimum time gap

The time gap to the vehicle ahead shall be at least 1.0 s in every gap setting.

ACC-R-013 Gap settings

The shortest selectable time gap setting shall be 0.8 s.

ACC-R-014 Set speed steps, long press

A long press on the speed button shall change the set speed in steps of 5 km/h.

ACC-R-015 Cut-in reaction

The system shall react fast to a vehicle cutting in.

ACC-R-016 Distance to stationary objects

The system shall keep a sufficient distance to stationary objects.

3 Longitudinal Dynamics

ACC-R-020 Takeover request timing

If the system reaches its limits, a takeover request shall be shown to the driver within 2 s.

ACC-R-021 Brake actuator

The system shall use the service brake for all decelerations.

ACC-R-022 Maximum deceleration

In ACC mode, the system shall decelerate the vehicle by at most 3.5 m/s².

ACC-R-024 Maximum acceleration

The longitudinal acceleration requested by ACC shall not exceed 2.5 m/s².

ACC-R-026 Activation speed

ACC shall only be active at vehicle speeds above 30 km/h.

ACC-R-027 Radar range

The radar sensor shall detect vehicles at a distance of at least 150 m.

ACC-R-028 Camera range

The camera shall detect lane markings at a distance of at least 60 m.

4 Comfort and Display

ACC-R-030 Following a braking vehicle

To follow a braking vehicle ahead, ACC may request a deceleration of up to 5 m/s².

ACC-R-031 Braking comfort

Braking interventions shall feel as smooth as possible.

ACC-R-032 System state display

The display shall inform the driver appropriately about the system state.

ACC-R-033 Set speed steps, short press

A short press on the speed button shall change the set speed in steps of 1 km/h.

ACC-R-034 Status indication

The ACC status shall be shown in the instrument cluster.

5 Driver Takeover and Degradation

ACC-R-041 Takeover request deadline

A takeover request shall be issued no later than 4 s after the system has reached its limits.

ACC-R-045 Acceleration after a stop

When resuming after a stop, ACC shall accelerate the vehicle with up to 0.3 g.

ACC-R-047 Stop-and-go operation

In stop-and-go traffic, ACC shall keep controlling the vehicle down to 5 m/s.

ACC-R-048 Weather robustness

The system shall work reliably in all weather conditions.

ACC-R-049 Fault recovery

After a sensor fault, the system shall recover within a reasonable time.

ACC-R-050 Switching off

The driver shall be able to switch ACC off with one button press.